

REQUIREMENT ANALYSIS

3.1 Customer journey map:

Ticket Created – Request logged, SLA assigned.

SLA Monitoring – Agent tracks SLA progress.

Risk Alert – Predicts possible SLA breach.

Virtual Agent Support – Explains ticket status.

Justification Capture – Collects delay reason.

Manager Approval – Sends justification for approval.

Customer Update – Shares delay and new ETA.

SLA Breach – Records breach if unavoidable.

Performance Review – Analyzes breaches for improvement.

	Customer Journey Map - Virtual Agent Driven SLA Breach Awareness & Justification System									
	Notice (Ticket Logged)		Map (SLA Attached)		Monitor (Monitoring & At-Risk Alert)		Breach Awareness & Justification (Breach & Justification)		Extend (Review & Improve)	
Steps	Agent logs incident ticket in the system		SLA auto-attaches based on category		Agent monitors status; Virtual Agent sends at-risk alert		SLA breaches; agent submits justification		Report reviewed in governance dashboard	
Interactions	Incident form	Service Desk portal	SLA engine	Business rules	Virtual Agent notification	SLA dashboard	Justification field	Approval workflow	SLA governance dashboard	Reports
Goals & Motivations	Log ticket quickly and accurately		Ensure correct SLA is applied		Resolve before breach / stay informed		Provide valid justification fast		Improve process to avoid repeat breaches	
Pain points	Missing/incorrect fields delay SLA attachment		Wrong business schedule causes incorrect timers		Alerts missed or delivered too late		Manual justification is time-consuming		No visibility into recurring breach reasons	
Negative moments	Ticket miscategorized, wrong SLA applied		SLA timer runs during non-working hours				Justification incomplete, ticket gets flagged		Repeated same-cause breaches go unnoticed	
Areas of opportunity	Auto-suggest category using AI		Auto-validate business schedule mapping		Proactive VA reminders before breach		VA-assisted guided justification entry		Root-cause analytics & dashboard alerts	

3.2 SOLUTION REQUIREMENT:

TEAM ID	SWTID-2026-9308
PROJECT NAME	VIRTUAL AGENT-DRIVEN SLA BREACH AWARENESS AND JUSTIFICATION SYSTEM

FUNCTIONAL REQUIREMENTS:

- Following are the functional requirements.

FR NO	Functional Requirement	Sub Requirement
FR-1	REQUIREMENT ANALYSIS & PLANNING	- Business Objectives - Functional Scope - Stakeholder Mapping - Execution Roadmap.
FR-2	BACKEND CONFIGURATION & DATA SETUP DATA ARCHITECTURE	- Update Set Creation - Create Business Schedule - SLA Creation - Configure SLA Start / Stop / Pause Conditions - Create Required Incident Fields - SLA Justification Fields
FR-3	NOTIFICATION CONFIGURATION USING FLOW DESIGNER & UI POLICY	- Create a Flow for Notification - UI Policy Configuration
FR-4	VIRTUAL AGENT DESIGN & CONFIGURATION	- Install Necessary Plugins - Configure Virtual Agent
FR-5	REPORTING & DASHBOARD SETUP	- Dashboard & Reports Navigation: - Adding reports to Existing Dashboard
FR-5	TESTING & VALIDATION	- Incident Creation - Test Case 1 - SLA Attachment & Running - Test case 2 - SLA At Risk Trigger - Test Case 3 - SLA Notification Email - Test Case 4 - UI Policy – Assignee Access - Test Case 5 - Virtual Agent Invocation - Test Case 6 - SLA Acknowledgement via Virtual Agent - Test Case 7 - Reporting – SLA Breaches by Reason - Test Case 8 - Reporting – Acknowledged vs Unacknowledged - Test Case 9 - Reporting – Pending Justifications - Test Case 10 - Dashboard – SLA Governance View - Test Case 11 - Conclusion

NON-FUNCTIONAL REQUIREMENTS:

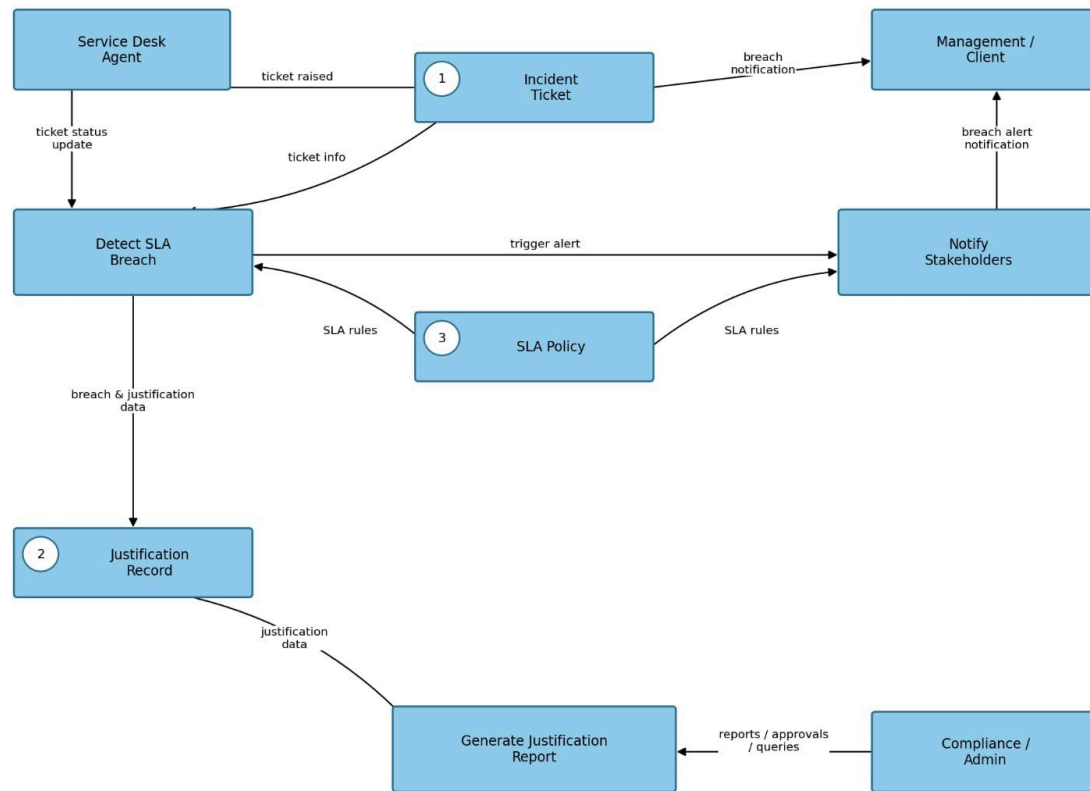
NFR NO.	Requirement	Description
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NFR-1	Usability	Easy-to-use interface for SLA alerts and justifications.
NFR-2	Security	Protects data using authentication and role-based access.
NFR-3	Reliability	Provides accurate SLA monitoring and notifications.
NFR-4	Performance	Sends real-time alerts with fast response times.
NFR-5	Availability	Available 24/7 with minimal downtime.
NFR-6	Scalability	Supports increasing users, incidents, and SLA records efficiently.

3.3 Data Flow Diagram:

A Data Flow Diagram (DFD) is a graphical representation that illustrates how data flows through The Virtual Agent–Driven SLA Breach Awareness & Justification System. It shows the movement Of data between users, the virtual agent, the SLA monitoring process, notifications, justification Generation, and the ServiceNow database. The DFD helps in understanding how the system Processes incident information, monitors SLA deadlines, sends proactive alerts, and generates SLA breach justifications.

Example: DFD Level 0 – Virtual Agent Driven SLA Breach Awareness & Justification System



3.4 Technology Stack

Component	Technology Used
Platform	ServiceNow
Frontend	ServiceNow Virtual Agent
Backend	ServiceNow Server-side Scripts
Database Custom Tables)	ServiceNow Tables (Task SLA, Incident,
Workflow	Flow Designer
Forms	Record Producer

Notifications

Email Notifications

Development Language

JavaScript, HTML, CSS

